

Extending Reti-Pioneer with Monotone Quality Routing and Endpoint-Aware Multi-Task Learning

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> One-sentence argument. Reti-Pioneer showed that frozen foundation features plus quality-aware fusion can screen systemic disease from color fundus photographs (CFPs); we keep that skeleton, replace fixed quality weights with a monotone bounded quality router, replace independent binary heads in the released training code with partial-label multi-task learning, and evaluate with an endpoint-aware cross-cohort protocol plus a calibration/decision-curve evaluation framework (not claimed as methodological novelty) on public cohorts reviewers can download – without inventing UK Biobank-comparable AUROCs from synthetic demo caches.

Abstract

Endocrine and metabolic disease screening from retinal imaging (oculomics) remains limited by image-quality heterogeneity, single-task training loops, and poorly calibrated risk scores. Reti-Pioneer (Zhang et al., *Nature Medicine*, 2026) demonstrated that frozen vision foundation models with quality-aware bilinear fusion can screen six systemic conditions on UK Biobank (UKB) and hospital cohorts. The published article describes a multitask screening framework; the released training code (main.py) trains per-disease independent binary loops. Quality routing is fixed at good=1 / usable=0.5 / bad=0. Here we propose a methods extension that (i) learns monotone bounded quality weights ($\text{bad} \leq \text{usable} \leq \text{good}$ in $[0, 1]$) as the default learnable path, (ii) shares a multi-label head with masked BCE over labels present for each sample/dataset, and (iii) reports expected calibration error (ECE), Brier score, and per-disease decision-curve analysis (DCA) under temperature scaling fitted on a fold disjoint from evaluation. We specify patient-level endpoint-aware cross-cohort evaluation on Ocular Disease Intelligent Recognition (ODIR-5K), Brazilian Multilabel Ophthalmological Dataset (BRSET), and Retinal Fundus Multi-disease Image Dataset (RFMiD), with explicit alignment flags (direct / partial / related_not_equivalent).

Honesty banner. Clinical AUROC / ECE / decision-curve tables are marked 待补充 until real images and foundation features replace synthetic caches. Pipeline sanity metrics on synthetic features appear only in the companion research report and are not manuscript claims. Calibration and DCA are an evaluation framework, not novelty claims. We do not invent UKB or hospital AUROCs.

Keywords: oculomics; fundus photography; monotone quality routing; multi-task learning; calibration (evaluation); endpoint harmonization; Reti-Pioneer

1. Introduction

Color fundus photography is widely available in community eye care and can encode microvascular signatures of systemic disease. Foundation models such as RETFound (Zhou et al., *Nature*, 2023) provide transferable retinal representations, while quality assessment resources such as EyeQ / MCF-Net (Fu et al., MICCAI 2019) support handling of imperfect acquisitions. Reti-Pioneer integrated frozen RETFound, Swin Transformer V2-B, and Vision Mamba-S with quality-aware fusion and metadata, reporting internal-test AUROCs of 0.699–0.833 across six diseases and

demonstrating a primary-care silent trial.

Despite this progress, three methodological gaps remain actionable without UKB access. First, Reti-Pioneer freezes quality weights to good=1 / usable=0.5 / bad=0; unconstrained learnable linear weights can violate clinical orderings, so we constrain the learnable path to be monotone and bounded. Second, although the Reti-Pioneer publication describes multitask screening, the released codebase trains independent binary heads in per-disease loops; a shared multi-label head with masked supervision better matches multi-label public ocular datasets. Third, screening deployment needs calibrated probabilities and decision-curve net benefit. We treat temperature scaling, ECE, Brier, and DCA as a standard evaluation framework rather than as the paper's methodological novelty. Population-independent multi-disease studies further show that single-site AUROC overstates transportability when cameras, taxonomies, and populations shift (e.g., Quellec et al., *Scientific Reports*, 2023).

Contributions.

1. A Reti-Pioneer-compatible monotone bounded quality router (default learnable path), with fixed 1/0.5/0 retained as the released-code baseline and unconstrained free-linear routing as an ablation only.

2. A /
sample

Boundary. We do not claim UKB type 2 diabetes mellitus (T2DM) AUROC 0.833 replication; we do not treat synthetic feature-cache AUROCs as clinical evidence; gout / osteoporosis / hyperlipidemia / thyroid remain UKB-only until public gold standards exist. ODIR diabetes (D) and BRSET diabetes labels are not UKB ICD endpoints. Over-broad diabetes_related mappings are exploratory only (related_not_equivalent).

2. Related work

Foundation models for CFP. RETFound and related retinal foundation models reduce labeled-data needs for ocular and oculomic tasks. Reti-Pioneer ensembles multiple frozen backbones rather than proposing a new foundation model; our work inherits that stance.

Quality-aware analysis. EyeQ-style classifiers produce good/usable/reject (bad) probabilities. Reti-Pioneer's bilinear quality fusion is a strong baseline. Monotone routing is a bounded refinement of the fusion weights, not a new quality taxonomy.

Multi-label fundus datasets and domain shift. ODIR-5K, RFMiD, and BRSET provide multi-label ocular (and some systemic) annotations downloadable by reviewers. They are not UKB ICD endpoints. We keep Reti-Pioneer's frozen-ensemble skeleton and change routing, head sharing, and endpoint-aware transfer reporting.

Calibration and decision curves (evaluation framework). Temperature scaling (Guo et al., ICML 2017), ECE, and DCA (Vickers & Elkin, 2006) are standard clinical-utility assessments. We adopt them as reporting protocol, not as claimed algorithmic novelty.

Positioning. This is a methods paper on top of Reti-Pioneer, not a competitor foundation model and not a *Nature Medicine*-scale clinical claim without biobank data.

3. Methods

3.1 Inputs

Paired or single CFPs, metadata (age, sex; weight/ethnicity padded when missing), and a 3-way quality probability vector. Patient IDs define folds so both eyes of one person stay in one split (`reti_pioneer.split`). RFMiD prefers official train/val/test CSVs when present. When `val_fraction` is set, we support train / calibration / val / test so temperature fitting never reuses the evaluation fold.

3.2 Frozen backbones and feature cache

Following Reti-Pioneer, we freeze RETFound / Swin V2-B / Vision Mamba-S and train on pre-extracted features (`fast_mode`). Public caches are built via `prepare_public_npz.py` then `extract_features.py`. Until real pixels and CUDA extraction succeed, caches may contain deterministic synthetic features flagged by `SYNTHETIC_FEATURES.txt` with `clinical_claim_allowed: false` – usable for continuous integration (CI) only.

3.3 Monotone bounded quality routing

QualityAware implements bilinear fusion with `quality_router` $\in$ {fixed, free_linear, monotone}:
| Mode | Behavior |

| fix

3.4 Partial-label multi-task head

ComplexModel supports `num_classes=K`. Masked BCE supervises only targets with label present (`target  $\geq$  0`); missing entries (`-1 / NaN`) are ignored. Released-code control = independent single-disease training loops as in upstream `main.py` (not a straw-man against the published “multitask” wording). Ensemble modes: `released_code` (train soft / eval max), `published_soft_vote`, `mean`, `temp_mean`. Legacy `ensemble="paper"` aliases to `released_code` with a deprecation warning.

3.5 Calibration and decision utility (evaluation framework)

Temperature `*T*` is fit on `calibration_ids` disjoint from `evaluation_ids`. If `--split val --calibrate` would fit and score the same set, `paper_mode` errors; otherwise we nest a held-out calibration slice and warn. We report AUROC / average precision (AP), equal-width ECE, Brier score, optional calibration intercept/slope, sensitivity at high specificity, per-disease DCA curves, and (illustrative only) net benefit at threshold 0.10 versus treat-all / treat-none.

3.6 Training protocol

| Item | Setting |

| Lea

3.7 Endpoint ontology (pre-registered)

Endpoint · Kind · ODIR · BRSET · Alignment notes

See docs/PUBLIC_DATA.md and reti_pioneer/label_map.py for the full harmonization table. Cross-eval with `--clinical-tables` / `--paper-mode` requires `alignment=direct`.

3.8 Failure modes and out-of-scope

- Missing quality vector → pad / default distribution (loader-defined).

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3.9 Reproducibility hooks

Configs: configs/ablation_e0.yaml ... e4.yaml (plus legacy ablation_*.yaml). Runner: scripts/run_ablations.py. Evaluation: scripts/evaluate.py (`--calibrate`, `--test-data-dir`, `--paper-mode`). Public checklist: python scripts/download_public_data.py.

4. Experiments (protocol)

- Datasets: ODIR-5K, BRSET, RFMiD (full-image download status: see Data availability; clinical tables 待补充).

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Honesty rule: never place synthetic/demo AUROC in submission tables.

5. Results

Clinical result tables: 待补充 (require real ODIR/BRSET/RFMiD pixels + foundation feature extraction on GPU).

For engineering verification only, a synthetic ablation matrix exists in results/ablation_summary.csv (companion report). Those values are pipeline sanity, typically near chance on tiny synthetic caches, and must not be compared to Reti-Pioneer's 0.699-0.833 UKB internal AUROCs.

Expected qualitative story after real features (hypothesis, not claim): modest AUROC gains from multi-task correlation; larger ECE reductions from temperature scaling on a disjoint calibration

fold; monotone routing helps more when true quality labels exist (BRSET); cross-cohort drop remains large and must be discussed under endpoint alignment constraints.

6. Discussion

Public ocular+systemic labels enable reproducible methods claims that UKB-gated papers cannot always support. Transferring Reti-Pioneer’s skeleton to ODIR/BRSET forces explicit endpoint harmonization and domain-shift reporting. Fairness and calibration should be first-class evaluation practices, matching the authors’ own limitations discussion – without overselling calibration/DCA as novelty.

Figure reading contract (companion report; manuscript clinical panels 待补充). Figure 1 is an information-flow schematic (yellow = monotone quality fusion; green = multi-task + calibration/DCA evaluation). Figures 2-4 currently plot SYNTHETIC feature-cache metrics near chance with tiny *n*; they justify engineering readiness and the *protocol* of reporting ECE and cross-cohort drop, not clinical superiority over Reti-Pioneer’s UKB AUROCs.

Honest novelty phrasing. Prefer: “we replace fixed quality weights with a monotone bounded router, train a masked multi-task head against the released independent-loop control, and evaluate with endpoint-aware cross-cohort splits plus a calibration/DCA reporting framework.” Avoid: “we outperform Reti-Pioneer” or “we achieve AUROC *X* on T2DM” without the original UKB endpoint and cohort.

Draft risks (pre-submission). (1) Results § still empty of real AUROC/ECE. (2) Published “multitask” vs released independent-head loops must stay explicit. (3) Endpoint drift if diabetes_related is misused clinically. (4) Monotone router without BRSET quality supervision may stay near (0, 0.5, 1). (5) DCA @ 0.10 is illustrative. (6) Without CUDA, foundation features remain 待补充.

7. Limitations

No UKB / SEED access in this workspace; label mismatch vs ICD systemic endpoints; no prospective trial; Vision Mamba optional on Windows; EyeQ weights external; synthetic caches for CI only; ODIR Kaggle credentials absent; BRSET PhysioNet DUA not completed here; GPU absent for RETFound/Swin extraction; E5 gating and patient-bootstrap CIs 待补充.

8. Reproducibility

Code, configs (configs/ablation_e*.yaml), scripts/run_ablations.py, patient-level split with calibration hashes, and docs/REPRODUCIBILITY.md.

Code availability (public snapshot for review):
<https://github.com/Coucou2016/retinal-imaging-methods-public> – code, configs, docs, and SYNTHETIC ablation CSV/figures only; no patient images or UKB extracts. See THIRD_PARTY_NOTICES.md. MIT LICENSE governs copyright; research disclaimers in README are not license restrictions.

Data availability: Public ODIR-5K / BRSET / RFMiD under their respective licenses (BRSET via PhysioNet credentialing). UK Biobank data are not redistributed here; clinical UKB AUROCs remain 待补充 until authorized access and re-analysis.

powershell

python

References

1. Zhang, X. et al. AI framework for multidisease detection via retinal imaging. **Nat Med** 32, 2494–2503 (2026). doi:10.1038/s41591-026-04359-w

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Assumptions or missing inputs

– Real ODIR/BRSET full-image downloads not completed (no Kaggle auth; BRSET PhysioNet credentialed).

– RFI
(CPU-

